P1 May 2011 TZ2

1.

1. A strange question.
Don't usually calculate speed from acceleration and time.
Does it mean final speed or average speed?
Lets assume final speed

$a = \frac{v-u}{t} = \frac{v}{t}$ $v = at$

$\Rightarrow$ add % error = 12% (B)

If on the other hand a had been calculated from v and t $a = \frac{v}{t}$

Then % error in a = % v + % t
$\Rightarrow$ % error in v = % a − % t
= 4%
Not an option so I guess its the first one

2.

2. Accel = gradient of v–t graph
∴ increasing accel = increasing grad.



(B)

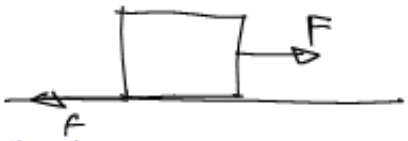
3.

3. Translational equil ⇒ resultant force = zero
- Not no external forces since forces could be balanced

(A)

4.

4.

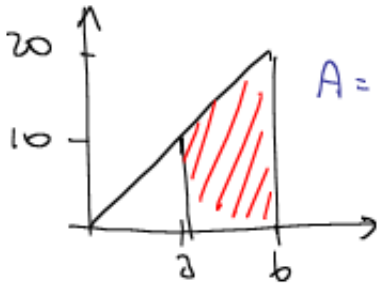


Resultant = F − f
rate of change of mom. = unbalanced F

(A)

5.

5. WD = area under graph



$A = \frac{1}{2}(20+10) \times 3$

$= 45\ \text{N cm}$

(C)

6.

6.

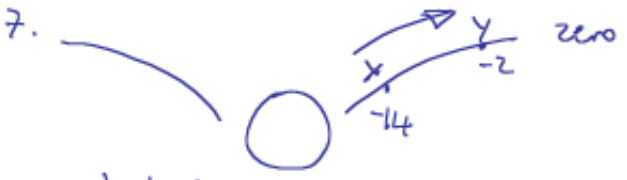


Path of projectile in uniform field is parabolic.

(A)

7.

7.



Potential is -ve and increasing to zero at ∞ $\Rightarrow$ moves away

$\Delta V = -2 - -14 = 12 \text{ J kg}^{-1}$

$\Rightarrow$ increase in V

C

8.

8.



$\frac{GMm}{r^2} = \frac{mv^2}{r}$

$v^2 = \frac{GM}{r}$

so bigger radius $\Rightarrow$ smaller v $\Rightarrow$ KE less

But work has been done $\Rightarrow$ total E is more

A

1

9.

9.

Energy of mol. of ideal gas is kinetic only.

D

10.

10.

Const. temp $\Rightarrow$ speed const.

increased vol $\Rightarrow$ collisions with walls less frequent

forces between mols of ideal gas = 0

$\Rightarrow$ I + II correct

B

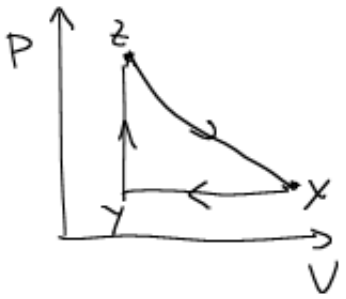
11.

11.
Adiabatic ⇒ work done = change in U
⇒ change in U = 50J (lost)
Const volume ⇒ no work done
so change in U = 20J (lost)
Total change in U = 70J

(A)

12.

12

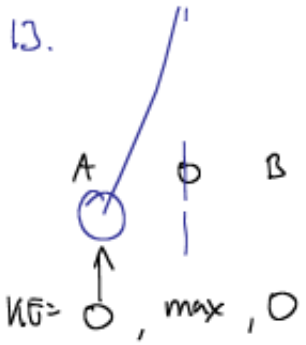


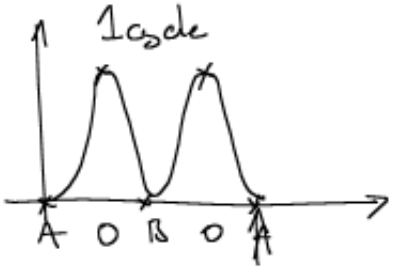
Work done on gas when gas compressed ⇒ X → Y

(D)

13.

13.





(C)

14.

14. $\phi = \frac{\text{path dif}}{\lambda} \times 2\pi = \frac{\lambda/4}{\lambda} \times 2\pi$

$= \frac{\pi}{2}$

C

15.

15. 4th harmonic $= 4f_1$

$\Rightarrow \lambda = \lambda_1/4$

D

Question should have referred to 1st Harmonic not fundamental.

16.

16. $\Delta f = \frac{v}{c} f_0$

car moves away $\Rightarrow$ f lower

car receives dopplar shifted signal and emits shifted signal

$\Rightarrow$ total shift $= \frac{2v}{c} f_0$

$\Rightarrow f' = f - \frac{2v}{c} f_0$

C

17.

17

An optically active liquid rotates plane of polarisation

D

18.

18.

Reducing λ will increase resolution
Increasing λ reduces resolution.
Think of blu ray

C

19.

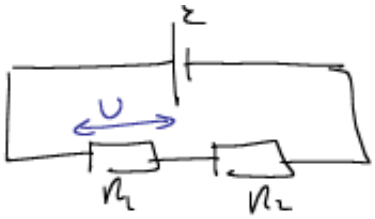
19.

Ampere = force between parallel
current carrying wires

B

20.

20.

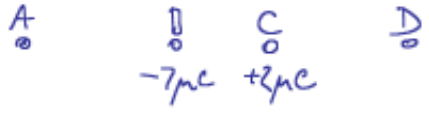


$I = \frac{\varepsilon}{R_1+R_2}$ $U = IR_1 = \frac{R_1}{R_1+R_2}\cdot\varepsilon$

A

21.

21



Zero point must be close to $+2\mu C$

D

22.

22.

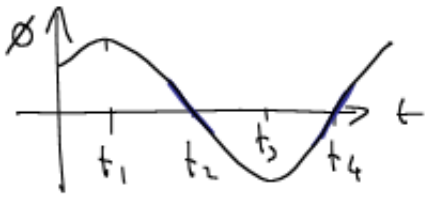


Gripnule

D

23.

23.



$\mathcal{E}$ ∝ rate of change of flux
– gradient of graph

⇒ max at t_2 + t_4

B

24.

24.
4A rms is current that gives same power as 4A dc

B

25.

25.

After 3T Activity of X = 1/8

Activity of Y = 1/2

$\frac{A_X}{A_Y} = \frac{1/8}{1/2} = \frac{2}{8} = \frac{1}{4}$

C

26.

26.

After fission total mass has reduced but BE has increased

D

27.

22.

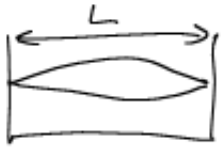
Increased intensity = more photons so rate of emission increases

KE depends on freq. so not changed

D

28.

28. $p = \frac{h}{\lambda}$



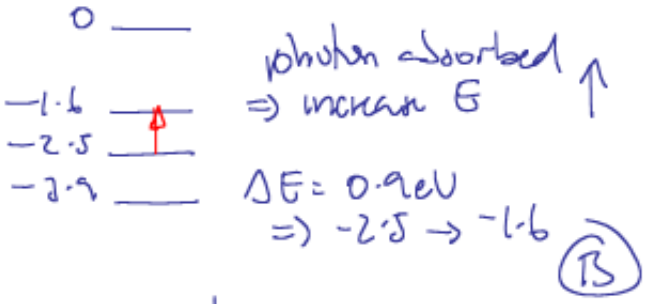
$\lambda = 2L$

$p = \frac{h}{2L}$

B

29.

29.



B

30.

30. $\Delta p \Delta x = \frac{h}{4\pi}$

Δx = delBroglie wavelength = $\frac{h}{p}$

$\Rightarrow \Delta p \times \frac{h}{p} = \frac{h}{4\pi}$ $\Delta p = \frac{p}{4\pi}$ (A)

31.

31.

Alpha & γ have discrete E levels

$\Rightarrow$ nuclear E levels

emission spectra $\Rightarrow$ e^- E levels

(C)

32.

32. $\frac{dN}{dt} = -\lambda N = A$

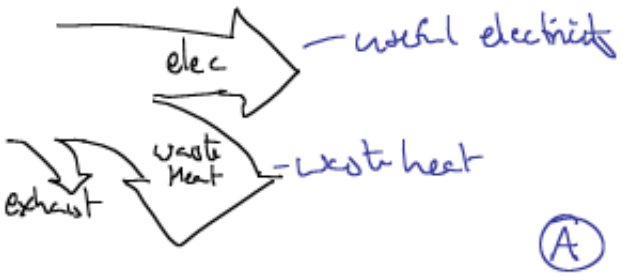
$N_S = \frac{-A}{\lambda_S}$ $N_T = \frac{-A}{\lambda_T}$

$\frac{N_S}{N_T} = \frac{\lambda_T}{\lambda_S}$ (A same for each)

(C)

33.

33.



(A)

34.

34

Coal – nuclear – geothermal

(D)

35.

35. $^{238}_{92}U$ $^{234}_{94}Pu$

U has 92 p^+ + 146 n°

Pu has 94 p^+ + 140 n°

⇒ must add 2 p^+ remove 1 n°

This can't be done in one step.

$^{238}_{92}U + n^\circ \rightarrow {}^{239}_{92}U \rightarrow {}^{239}_{93}Np + e^- + \bar{\nu}_e$

$^{239}_{93}Np \rightarrow {}^{239}_{-4}P_u + e^- + \bar{\nu}$

(A)

36.

36. $PE = mgh = \rho A h g h$

$PE \propto h^2$

(C)

37.

37.

Atmosphere absorbs IR radiation

(others are false)

(B)

38.

38

16	8	4	2	1
1	1	0	0	1

16 + 8 + 1 = 25

(C)

39.

39. Linear mag $= \frac{h_i}{h_o}$



$M = \sqrt{\frac{50}{2}} \times 10^{-3}$

$= \sqrt{25} \times 10^{-3} = 5 \times 10^{-3}$

(B)

40.

40.

Samples in time $T = \frac{T}{t}$

each sample has n bits

$\Rightarrow n\frac{T}{t}$ bits

(A)

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